

KUQA, China

WMO: 516440

Lat: 41.717N

Lon: 82.950E

Elev: 3555

StdP: 12.90

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	2.2	5.9	-4.8	4.8	13.7	-1.7	5.7	17.0	9.4	26.5	7.8	24.3	2.6	0	0.440

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.8	93.5	64.7	91.2	63.9	89.1	63.2	68.2	85.5	66.8	84.4	65.6	83.3	5.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.1	98.9	75.2	61.2	92.1	74.2	59.4	86.4	73.3	34.8	85.4	33.6	84.2	32.5	83.5	76.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
14.2	11.6	9.7	DB	0.7	98.2	5.0	2.4	-2.9	99.9	-5.8	101.3	-8.5	102.6	-12.1	104.3	(4)
			WB	0.0	71.9	4.9	1.9	-3.5	73.3	-6.4	74.5	-9.1	75.5	-12.6	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	52.7	18.8	30.8	46.2	59.5	67.7	74.2	76.9	74.9	66.3	53.6	38.7	23.1
	DBStd	20.77	5.74	7.00	7.18	7.02	6.09	5.00	4.38	4.70	4.91	6.25	6.74	5.83
	HDD50	2885	966	537	162	12	0	0	0	0	35	340	833	
	HDD65	5689	1431	957	581	188	43	3	0	1	42	356	789	1298
	CDD50	3859	0	0	45	296	549	726	834	772	489	146	2	0
	CDD65	1187	0	0	0	22	127	279	369	307	81	2	0	0
	CDH74	12920	0	0	2	294	1324	3001	4088	3290	875	46	0	0
	CDH80	5084	0	0	0	45	375	1225	1870	1360	206	3	0	0
Wind	WSAvg	4.2	3.0	3.9	4.7	5.2	5.2	5.0	4.8	4.6	4.1	3.6	3.4	3.0
Precipitation	PrecAvg	2.4	0.0	0.1	0.1	0.1	0.2	0.5	0.5	0.4	0.2	0.2	0.1	0.0
	PrecMax	4.9	0.2	0.6	0.6	0.6	1.1	2.6	2.0	1.5	1.1	1.1	1.1	0.3
	PrecMin	1.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.9	0.1	0.1	0.2	0.2	0.2	0.6	0.4	0.4	0.2	0.3	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	52.9	72.5	84.4	89.9	94.2	97.9	95.4	88.4	78.9	63.5	42.5
		MCWB	29.8	38.5	48.3	56.1	60.7	63.5	66.5	66.0	62.8	55.6	45.8	34.1
	2%	DB	34.4	48.7	67.7	80.6	86.9	91.8	94.6	92.0	84.9	74.4	59.1	38.2
		MCWB	27.2	36.5	46.0	54.3	59.4	62.6	65.6	65.0	60.9	54.0	43.0	31.1
	5%	DB	31.3	45.7	64.2	77.6	84.2	89.6	91.7	89.8	82.2	70.8	55.3	35.1
		MCWB	25.3	34.6	44.4	52.7	57.9	62.0	64.8	64.3	60.1	51.9	41.9	28.9
	10%	DB	28.7	42.3	60.2	73.9	81.5	87.2	89.2	87.5	79.3	67.2	51.0	32.2
		MCWB	23.7	32.4	42.6	51.0	56.7	61.3	63.9	63.9	59.0	50.2	39.9	26.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.2	39.4	49.0	57.4	63.6	67.5	71.1	70.2	65.9	57.2	46.6	35.0
		MCDB	38.0	50.9	70.2	82.0	83.5	84.8	87.2	86.0	83.1	74.3	61.6	42.0
	2%	WB	27.6	36.9	46.7	55.3	61.0	65.5	68.6	68.1	63.3	55.4	43.9	31.4
		MCDB	33.6	47.7	65.6	77.8	83.2	84.1	86.0	85.1	79.8	71.3	57.1	37.4
	5%	WB	25.6	35.1	44.9	53.6	59.3	64.1	67.2	66.7	61.8	53.5	42.1	29.2
		MCDB	31.1	44.8	62.6	74.8	81.0	83.5	85.0	84.0	78.0	67.9	54.2	34.7
	10%	WB	23.9	33.0	43.1	51.9	57.8	62.9	65.9	65.6	60.3	51.6	40.2	27.1
		MCDB	28.7	41.4	59.3	72.1	78.6	82.5	83.9	82.8	76.0	64.1	50.5	31.8
Mean Daily Temperature Range	5% DB	MDBR	18.3	18.6	20.1	21.0	21.8	21.9	21.8	22.2	23.5	25.2	20.5	16.4
		MCDBR	20.3	21.3	24.8	26.8	26.2	25.4	25.9	25.6	27.4	28.3	25.8	18.5
		MCWBR	14.6	11.9	11.8	10.4	10.0	8.9	8.9	9.3	10.0	11.0	13.2	12.5
	5% WB	MCDBR	19.8	20.2	23.7	25.2	23.7	22.6	22.7	22.8	25.1	26.8	25.3	17.9
		MCWBR	14.4	11.4	11.3	10.0	10.2	9.0	9.1	9.6	10.1	10.4	13.1	12.1
Clear-Sky Solar Irradiance	taub	0.336	0.397	0.492	0.543	0.556	0.576	0.536	0.571	0.579	0.490	0.399	0.353	
	taud	2.119	1.983	1.765	1.684	1.713	1.718	1.837	1.745	1.685	1.805	1.979	2.062	
	Ebn at Noon	246	246	234	233	235	230	239	224	207	210	216	225	
	Edh at Noon	33	45	63	74	74	73	64	69	68	54	38	33	
All-Sky Solar Radiation	RadAvg	737	1048	1397	1726	1959	2021	1986	1794	1528	1185	811	628	
	RadStd	65	62	65	99	92	106	96	86	60	54	55	61	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	-0.79	N/A	N/A	N/A	N/A	-92
Regional (2 neighbors)		N/A	N/A	N/A	N/A	-0.90	N/A	N/A	N/A	N/A	-92

Nomenclature: See separate page